# Simple CMOS Process

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|---|---|---|---|---|
| 9. | Pwell oxidation<br>（40±5nm）<br>2-4# PADOXTST | RCA Clean<br>Oxidation into furnace<br>Oxidation outof furnace | | 1. Thickness of SiO2 inside Nwell<br>2. Thickness of SiO2 outside Nwell |
| 10. | Pwell Photolithography<br><br>mask: PWELL | Pretreatment<br>Photo deposit<br>Exposure<br>Development<br>Check | In simulation：<br>1. Deposit photo 1.2um<br>2. Etch photo with the correct shape<br>(shape for Pwell should be obtained in the 2D profile of the device) | Plot the 2D str picture to check the etch of photo |
| 11. | P well implant | P well implant | B 110Kev 5E12<br>If you use deckbuild menu to generate the codes, pls. Delete the definition for tilt and rotation | 1D curve of Boron concentration (curve 1), x axis for 1D curve range 0-10 |
| 12. | Remove all of the photoresist | Etch photo | | |
| 13. | Drive-in Pwell<br><br>3-3# PWELL | 3# Clean<br>Oxidation into furnace<br>Oxidation outof furnace | | Thickness of SiO2 x=3.6<br>1D curve of Boron concentration (curve 2), plot curve1 and 2 in one picture<br>x axis range 0-10 |

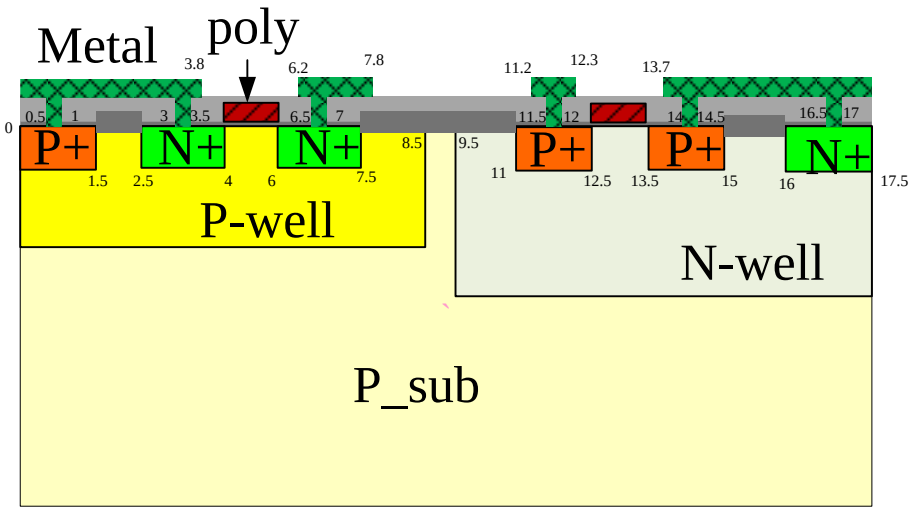
| | | | | |
|---|---|---|---|---|
| 14. | Remove all of SiO2 | Wet etch SiO2 | | |
| 15. | Active region oxidation<br>(40±5nm)<br>2-4# PADOXTST | RCA Clean<br>Oxidation into furnace<br>Oxidation outof furnace | | Thickess of SiO2 in the active region |
| 16. | SiN deposit | 3# Clean<br>CVD into furnace<br>CVD into furnace | 150nm div=5<br>SiN material is nitride<br>In simulation, just use deposit (No CVD) | |
| 17. | Active Photolithography<br>Mask: ACTIVE | Pretreatment<br>Photo deposit<br>Exposure<br>Development<br>Check | （No simulation）<br>Consider:<br>Why there is no need for this step simulation? | |
| 18. | Etch SiN<br>仿真内容：<br>根据 Active 形貌刻蚀 SiN | | According to the shape of Active to etch SiN<br>Note：SiN remain region is Active region<br>Active is the thin oxide region (dark grey) | |
| 19. | Remove all of photo | Wet etch to remove photo<br>check | （No simulation）<br>Because step 17 is not simulated | |

# Simple CMOS Process

oxidation 2-4#

| PADOX1 2-4 400±50A | | | | | | | |
|---|---|---|---|---|---|---|---|
| Step | Temp °C | Time | N2(slpm) | O2(slpm) | H2(slpm) | Tca (sccm) | Boat Cm/min |
| Push in | 800 | 34’ | 8 | | | | |
| Ramp to | 1000 | 25’ | 8 | | | | |
| Sta temp | 1000 | 5’ | 8 | | | | |
| Tca dryox | 1000 | 45’30” | | 8 | | | |
| Ramp to | 900 | 20’ | 8 | | | | |
| Anneal | 900 | 30’ | 8 | | | | |
| Ramp to | 800 | 1h | 6.8 | | | | |
| Push out | 800 | 15’ | 6.8 | | | | |

| PWELL 3-3# drive-in Pwell | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Step | Temp °C | Ramp °C/min | Time | N2(slpm) | O2(slpm) | H2(slpm) | Tca (sccm) | Boat Cm/min |
| Push in | 800 | | 20’ | | 8 | | | |
| Ramp to | 1130 | | 60’ | | 8 | | | |
| Anneal | 1130 | | 210’ | 8 | 0.6 | | | |
| Ramp to | 800 | | 120’ | 8 | | | | |
| Push out | 800 | | 15’ | 6 | | | | |




NWELL PWELL ACTIVE NPLUS PPLUS POLY CONT METAL